

# **SafeStep: A Low-Cost IoT-Enabled Semi-Autonomous Smart Walker for Blind and Elderly Navigation with Obstacle Avoidance, Water Hazard Detection, and Emergency Assistance**

Submitted in Partial Fulfillment of the Requirements for the Degree of  
Bachelor of Science in Computer Science and Engineering of the  
University of Asia Pacific

by

**MD. Sydul Islam Zubair**

Roll: 22201174

**Suna Hera Akter Suna**

Roll: 22201168

**Farhana Haque Mahima**

Roll: 22201175

Supervised by:

**Nazmun Nahid**

Assistant Professor



Department of Computer Science & Engineering  
University of Asia Pacific

June 2026

## Declaration

We, hereby, declare that the work presented in this Thesis is the outcome of the investigation performed by us under the supervision of Nazmun Nahid, Assistant Professor, Department of Computer Science, University of Asia Pacific. We also declare that no part of this Thesis and thereof has been or is being submitted elsewhere for the award of any degree or Diploma.

### Countersigned

---

(Nazmun Nahid)  
Supervisor

### Signature

.....  
MD. Sydul Islam Zubair

.....  
Suna Hera Akter Suna

.....  
Farhana Haque Mahima

## Certificate of Approval

We hereby recommend that the thesis prepared by [name of the group members] entitled “[thesis/project title here]” is accepted as fulfilling the part of the requirements for degree of Bachelor of Science in Computer Science and Engineering.

.....

Name of the Supervisor

Chairman of the Committee

Designation of the Supervisor

(Supervisor)

Department of Computer Science and Engineering

University of Asia Pacific (UAP)

.....

Name of the Examiner

Member of the Committee

Designation of the Examiner

(External)

Department of Computer Science and Engineering

University of Asia Pacific (UAP)

.....

Name of the Head of the Dept

Head of the Department

Designation of the Head of the Dept

Department of Computer Science and Engineering

University of Asia Pacific (UAP)

## Acknowledgements

First of all I would like to thank the almighty ALLAH. Today I am successful in completing my work with such ease because He gave me the ability, chance, and cooperating supervisor.

I would like to take the opportunity to express my gratitude to Nazmun Nahid, my respected supervisor. Although she was always loaded with several other activities, she gave me more than enough time in this work. She not only gave me time but also proper guidance and valuable advice whenever I faced with some difficulties. Her comments and guidance helped me in preparing my thesis report.

Also I am grateful to my family; who are, always with me in my every step of life.

# Abstract

The demand for affordable assistive navigation technologies is rising every single day due to the growth in elderly population and the various problem / challenges faced by the visually impaired people. Also the traditional solutions such as walking sticks or walkers provide very few autonomous or semiautonomous mechanisms for obstacle avoidance and emergency communication. So, as a solution we suggest our project SafeStep which is a IoT-enabled semi-autonomous smart walker aimed to prioritize navigation independence for blind or elderly people. The SafeStep system uses an ESP32 microcontroller, three ultrasonic sensors, a water detection sensor, dual buzzer feedback mechanisms, DC motor-assisted navigation, and cloud-based emergency communication through the Blynk IoT platform. The main functionalities of the walker may be summarized as continuously monitoring its surroundings, detecting obstacles in multiple directions, identifying wet or slippery surfaces and providing directional guidance using sound alerts. Moreover, the emergency SOS feature allows the user to instantly send notification to his or her caregiver. A prototype was built and was tested in various environments to test its ability in obstacle detection, water hazard detection, navigation assistance, and emergency notifications. And the results showed that SafeStep prototype approximately correctly detected 90% of the obstacles correctly within the range of 5 cm to 150 cm and the optimal performance was observed between 20 cm to 80 cm. SafeStep was able to reduce the response time below one second and while delivering SOS notifications the response time was approximately one to two seconds. The proposed solution showed that embedded systems and IoT technologies can be effectively used to provide practical mobility assistance for elderly and visually impaired individuals at low-cost. SafeStep aims to work as a scalable foundation for the future of assistive robots in Bangladesh.

**Keywords:** Smart Walker, Assistive Robotics, IoT, ESP32, Obstacle Detection, Elderly Assistance, Blind Navigation, Ultrasonic Sensors, Emergency Notification, Embedded Systems

# Contents

|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| <b>List of Figures</b>                                          | <b>ix</b> |
| <b>List of Tables</b>                                           | <b>x</b>  |
| <b>1 Introduction</b>                                           | <b>1</b>  |
| 1.1 Background and Motivation . . . . .                         | 1         |
| 1.2 Problem Statement . . . . .                                 | 2         |
| 1.3 Objectives of the Project . . . . .                         | 2         |
| 1.4 Research Questions / Hypotheses . . . . .                   | 3         |
| 1.5 Scope of the Work . . . . .                                 | 3         |
| 1.6 Expected Outcomes . . . . .                                 | 4         |
| 1.7 Impacts of the Project . . . . .                            | 4         |
| 1.7.1 Impact on Societal and Cultural Issues . . . . .          | 4         |
| 1.7.2 Impact on Health, Safety, and Legal Issues . . . . .      | 4         |
| 1.7.3 Impact on Environment and Sustainability Issues . . . . . | 5         |
| 1.8 Report Organization . . . . .                               | 5         |
| <b>2 Background and Literature Review</b>                       | <b>6</b>  |

|          |                                                          |           |
|----------|----------------------------------------------------------|-----------|
| 2.1      | Preliminaries . . . . .                                  | 6         |
| 2.1.1    | Internet of Things (IoT) . . . . .                       | 6         |
| 2.1.2    | Ultrasonic Distance Measurement . . . . .                | 6         |
| 2.1.3    | Assistive Navigation Systems . . . . .                   | 7         |
| 2.1.4    | Embedded Systems . . . . .                               | 7         |
| 2.1.5    | Blynk IoT Platform . . . . .                             | 7         |
| 2.2      | Related Works / Existing Systems . . . . .               | 7         |
| 2.3      | Comparative Analysis of Related Works . . . . .          | 8         |
| 2.4      | Research Gap / Limitations of Existing Methods . . . . . | 9         |
| 2.5      | Summary . . . . .                                        | 9         |
| <b>3</b> | <b>System Analysis and Design</b>                        | <b>10</b> |
| 3.1      | Requirement Analysis . . . . .                           | 10        |
| 3.1.1    | Stakeholder Analysis . . . . .                           | 10        |
| 3.1.2    | Functional Requirements . . . . .                        | 11        |
| 3.1.3    | Non-Functional Requirements . . . . .                    | 12        |
| 3.2      | System Overview / Architecture . . . . .                 | 12        |
| 3.2.1    | High-Level Architecture . . . . .                        | 13        |
| 3.2.2    | Detailed Design Diagrams . . . . .                       | 14        |
| 3.2.3    | Design Alternatives and Rationale . . . . .              | 14        |
| 3.3      | Algorithmic Flow and Pseudocode . . . . .                | 16        |
| 3.4      | Summary . . . . .                                        | 19        |

|          |                                                       |           |
|----------|-------------------------------------------------------|-----------|
| <b>4</b> | <b>Methodology and Implementation</b>                 | <b>20</b> |
| 4.1      | Methodological Framework . . . . .                    | 20        |
| 4.2      | Tools, Libraries, and Technologies Used . . . . .     | 21        |
| 4.3      | Data Collection / Dataset Description . . . . .       | 21        |
| 4.4      | Preprocessing Steps . . . . .                         | 22        |
| 4.5      | Model Development / Algorithm Design . . . . .        | 22        |
| 4.6      | System Implementation / Modules Description . . . . . | 23        |
| 4.7      | Hardware and Software Requirements . . . . .          | 24        |
| 4.8      | Summary . . . . .                                     | 26        |
| <b>5</b> | <b>Results and Evaluation</b>                         | <b>27</b> |
| 5.1      | Experimental Setup . . . . .                          | 27        |
| 5.1.1    | Testing Scenarios . . . . .                           | 28        |
| 5.2      | Performance Metrics . . . . .                         | 28        |
| 5.3      | Result Analysis . . . . .                             | 29        |
| 5.3.1    | Obstacle Detection Performance . . . . .              | 29        |
| 5.3.2    | Water Hazard Detection Performance . . . . .          | 30        |
| 5.3.3    | SOS Notification Performance . . . . .                | 30        |
| 5.3.4    | Navigation Performance . . . . .                      | 31        |
| 5.3.5    | Battery Performance . . . . .                         | 32        |
| 5.4      | Comparison with Existing Approaches . . . . .         | 32        |
| 5.5      | Discussion on Findings . . . . .                      | 33        |



|          |                                                               |           |
|----------|---------------------------------------------------------------|-----------|
| 5.6      | Limitations . . . . .                                         | 33        |
| 5.7      | Summary . . . . .                                             | 33        |
| <b>6</b> | <b>Project Planning and Budget</b>                            | <b>35</b> |
| 6.1      | Project Timeline . . . . .                                    | 35        |
| 6.2      | Budget Estimation . . . . .                                   | 36        |
| 6.2.1    | Cost-Benefit Analysis . . . . .                               | 37        |
| 6.3      | Summary . . . . .                                             | 38        |
| <b>7</b> | <b>Standards and Design Constraints</b>                       | <b>39</b> |
| 7.1      | Compliance with Standards and Professional Practice . . . . . | 39        |
| 7.1.1    | Software Standard . . . . .                                   | 39        |
| 7.1.2    | Hardware Standard . . . . .                                   | 40        |
| 7.1.3    | Assistive Technology Considerations . . . . .                 | 40        |
| 7.2      | Design Constraints . . . . .                                  | 40        |
| 7.2.1    | Ethical and Professional Responsibilities . . . . .           | 41        |
| 7.2.2    | Environmental and Sustainability Considerations . . . . .     | 42        |
| 7.2.3    | Health and Safety Considerations . . . . .                    | 42        |
| 7.3      | Complex Engineering Problem (CEP) Coverage . . . . .          | 43        |
| 7.4      | Complex Engineering Activities (CEA) Coverage . . . . .       | 44        |
| 7.5      | Summary . . . . .                                             | 45        |
| <b>8</b> | <b>Conclusion and Future Work</b>                             | <b>46</b> |

|                   |                                            |           |
|-------------------|--------------------------------------------|-----------|
| 8.1               | Summary of the Work . . . . .              | 46        |
| 8.2               | Key Findings and Contributions . . . . .   | 46        |
| 8.3               | Limitations of the Study . . . . .         | 47        |
| 8.4               | Recommendations and Future Work . . . . .  | 48        |
| <b>Appendices</b> |                                            | <b>51</b> |
| A                 | Source Code / GitHub Link . . . . .        | 51        |
| B                 | Team Responsibilities . . . . .            | 51        |
| C                 | Ethics and Compliance Statements . . . . . | 52        |
| D                 | Survey 1 . . . . .                         | 52        |
| E                 | Survey 2 . . . . .                         | 53        |

# List of Figures

|     |                                                            |    |
|-----|------------------------------------------------------------|----|
| 3.1 | SafeStep Control System Diagram . . . . .                  | 13 |
| 3.2 | Hardware Achitecture of SafeStep Embedded System . . . . . | 14 |
| 3.3 | Circuit Diagram . . . . .                                  | 14 |
| 3.4 | SafeStep Flowchart . . . . .                               | 16 |
| 4.1 | Project Development Framework . . . . .                    | 20 |
| 4.2 | Blynk Communication Workflow . . . . .                     | 24 |
| 4.3 | Prototype Photograph . . . . .                             | 26 |
| 5.1 | Obstacle Detection Accuracy Chart . . . . .                | 29 |
| 5.2 | Water Detection Reliability . . . . .                      | 30 |
| 1   | Collected Images while conducting survey 2 . . . . .       | 55 |

# List of Tables

|     |                                                                     |    |
|-----|---------------------------------------------------------------------|----|
| 2.1 | Comparative Analysis of Existing Assistive Navigation Systems . . . | 8  |
| 3.1 | Functional Requirements . . . . .                                   | 11 |
| 3.2 | Non-Functional Requirements . . . . .                               | 12 |
| 3.3 | Obstacle Detection Technologies . . . . .                           | 15 |
| 3.4 | Controller Selection . . . . .                                      | 15 |
| 4.1 | Development Technologies . . . . .                                  | 21 |
| 4.2 | Libraries Used . . . . .                                            | 21 |
| 4.3 | Threshold Values . . . . .                                          | 22 |
| 4.4 | Hardware Components . . . . .                                       | 24 |
| 4.5 | Software Requirements . . . . .                                     | 25 |
| 5.1 | Test Scenarios . . . . .                                            | 28 |
| 5.2 | Obstacle Detection Results . . . . .                                | 29 |
| 5.3 | Water Detection Results . . . . .                                   | 30 |
| 5.4 | SOS Notification Results . . . . .                                  | 31 |
| 5.5 | Navigation Evaluation . . . . .                                     | 31 |
| 5.6 | Battery Analysis . . . . .                                          | 32 |
| 5.7 | Comparative Analysis . . . . .                                      | 32 |
| 6.1 | Seven-Week Project Timeline . . . . .                               | 36 |
| 6.2 | Hardware Cost Breakdown . . . . .                                   | 37 |
| 6.3 | Cost Distribution . . . . .                                         | 37 |
| 7.1 | CEP Mapping . . . . .                                               | 43 |
| 7.2 | CEA Mapping . . . . .                                               | 44 |

# Chapter 1

## Introduction

### 1.1 Background and Motivation

Mobility plays a fundamental role in maintaining independence, confidence, and quality of life for both elderly individuals and people with visual impairments. According to global healthcare studies, falls and navigation-related accidents remain among the leading causes of injury for older adults and visually impaired populations. Traditional mobility aids such as walking sticks and conventional walkers provide physical support but offer limited environmental awareness and no active hazard detection capabilities.

Recent advancements in embedded systems, Internet of Things (IoT), robotics, and sensor technologies have enabled the development of intelligent assistive devices capable of providing real-time environmental awareness and user assistance. Smart mobility aids can enhance navigation by detecting obstacles, identifying hazards, and communicating emergency situations to caregivers or family members.

Despite significant progress in assistive technology, many commercially available smart walkers remain expensive and inaccessible in developing countries such as Bangladesh. Furthermore, existing solutions often focus on a single functionality such as obstacle detection or GPS tracking, while overlooking additional safety concerns including wet surface detection and emergency communication.

To address these limitations, this project proposes SafeStep, a low-cost IoT-enabled

semi-autonomous smart walker designed to assist blind and elderly individuals through obstacle avoidance, water hazard detection, directional audio guidance, and emergency SOS notification capabilities.

## **1.2 Problem Statement**

Various navigation related challenges both in indoor and outdoor environment is faced by the blind and elderly people in our country. Traditional method lack the ability to detect obstacles before any sort of physical contact which increases the possibility of collisions, falls and injuries.

Many existing smart assistance systems have tried to address this problem through obstacle detection mechanisms but most of these solutions have one or more of the following limitations: High implementation cost, sensing coverage limits, lackings in water hazard detection, absence of emergency communication features, complex architectures unsuitable for developing regions and lastly high power consumption.

The absence of a low-cost, multi-functional, and easily deployable assistive navigation system motivates the development of SafeStep.

## **1.3 Objectives of the Project**

The main objective of our project is to design and implement a low-cost effective smart walker that increases the safety and navigational independence of elderly and visually impaired individuals.

The more specific objectives of our goals would be to design a semi-autonomous smart walker prototype, to detect obstacles using multiple ultrasonic sensors, to allow autonomous directional navigation, to detect wet or slippery surfaces using a water sensor, to provide audio directional feedback using buzzers, to add a IoT-based emergency notification mechanism, to develop a low-cost solution suitable for developing countries such as Bangladesh and lastly to evaluate system performance under real-world operating conditions.

## 1.4 Research Questions / Hypotheses

The project is guided by the following research questions:

**RQ1:** Can a low-cost smart walker provide reliable obstacle detection for blind and elderly users?

**RQ2:** Can ultrasonic sensor-based navigation assistance improve user safety during movement?

**RQ3:** How effectively can water hazard detection reduce environmental risks?

**RQ4:** Can IoT-based emergency notification improve caregiver awareness during emergencies?

**RQ5:** Is it possible to develop an affordable assistive navigation platform suitable for deployment in resource-constrained environments?

## 1.5 Scope of the Work

So we can clearly identify the scope of this project to :

- Design and construction of a semi-autonomous walker prototype.
- Integration of ESP32 microcontroller.
- Obstacle detection using three HC-SR04 ultrasonic sensors.
- Water hazard detection using a water sensor.
- Audio alert generation through dual buzzers.
- IoT integration through Blynk cloud services.
- Emergency SOS communication mechanism.
- Experimental testing and performance evaluation.

And lastly we want to highlight the fact that the project does not include: Computer vision systems nor GPS navigation nor AI-based object classification nor Medical monitoring systems and lastly we may include that this project is not a Commercial-grade mechanical construction.

## **1.6 Expected Outcomes**

By the end of our project we aim to successfully develop a functional smart walker prototype which will have reliable obstacle detection capability along with real-time water hazard detection, effective directional guidance system, cloud-based emergency alert mechanism and improved user safety and mobility assistance which will lead to a successful demonstration of a scalable assistive robotics platform.

## **1.7 Impacts of the Project**

### **1.7.1 Impact on Societal and Cultural Issues**

The proposed system aims to improve accessibility and independence for vulnerable populations including elderly individuals and visually impaired users. Increased mobility can contribute to improved social participation, confidence, and quality of life.

### **1.7.2 Impact on Health, Safety, and Legal Issues**

The system contributes to fall prevention through proactive obstacle and hazard detection. Emergency notification functionality enables timely communication with caregivers during potentially dangerous situations.



### 1.7.3 Impact on Environment and Sustainability Issues

SafeStep utilizes low-power embedded hardware and inexpensive electronic components. The system demonstrates efficient utilization of computational resources while maintaining affordability and sustainability.

## 1.8 Report Organization

This report is organized into eight chapters.

**Chapter 1** introduces the research problem, objectives, scope, and motivation.

**Chapter 2** presents theoretical background and literature review.

**Chapter 3** discusses system analysis, architecture, and design considerations.

**Chapter 4** describes methodology, implementation, and technical development.

**Chapter 5** presents experimental evaluation and performance analysis.

**Chapter 6** discusses project planning and budget estimation.

**Chapter 7** examines standards compliance, design constraints, and engineering accreditation requirements.

**Chapter 8** concludes the study and outlines future research opportunities.

# Chapter 2

## Background and Literature Review

### 2.1 Preliminaries

Assistive robotics is an interdisciplinary field that combines robotics, embedded systems, sensing technologies, and intelligent control mechanisms to improve the quality of life of individuals with physical or sensory impairments.

#### 2.1.1 Internet of Things (IoT)

The Internet of Things refers to interconnected physical devices capable of sensing, processing, and exchanging information through communication networks.

#### 2.1.2 Ultrasonic Distance Measurement

Ultrasonic sensors measure distance by transmitting high-frequency sound waves and calculating the time required for reflected signals to return.

The distance calculation is given by:

$$Distance = \frac{(Speed of Sound \times Time)}{2} \quad (2.1)$$

### **2.1.3 Assistive Navigation Systems**

Assistive navigation systems help users avoid hazards and navigate safely using environmental sensing technologies.

### **2.1.4 Embedded Systems**

An embedded system combines hardware and software components designed for dedicated functionalities within larger systems.

### **2.1.5 Blynk IoT Platform**

Blynk is a cloud-based IoT platform that enables remote monitoring, control, and event notification for embedded devices.

## **2.2 Related Works / Existing Systems**

Numerous researchers have proposed intelligent mobility assistance systems for visually impaired and elderly individuals.

Recent studies have demonstrated the effectiveness of ultrasonic sensors in obstacle detection applications. Several smart cane systems utilize single or dual ultrasonic sensors to provide warning signals to users. While these systems successfully identify obstacles, they often lack autonomous navigation support.

Research conducted on IoT-enabled smart sticks introduced wireless communication features for emergency assistance. These systems improved caregiver awareness but generally focused on user notification rather than active navigation support.

Several autonomous walker systems have incorporated advanced technologies such as LiDAR, cameras, and machine learning algorithms. Although these solutions provide high accuracy, their implementation costs are often prohibitively expensive for deployment in developing countries.

Recent literature also highlights increasing interest in integrating hazard detection capabilities such as wet floor recognition. However, these systems are typically implemented separately from navigation assistance platforms.

The review reveals that most existing solutions address only one or two assistive functionalities rather than providing a unified safety framework.

## 2.3 Comparative Analysis of Related Works

Table 2.1: Comparative Analysis of Existing Assistive Navigation Systems

| Study                                                                                                           | Obstacle<br>Detection | Water<br>Detection | Emerger<br>Alert | IoT<br>Support | Cost   |
|-----------------------------------------------------------------------------------------------------------------|-----------------------|--------------------|------------------|----------------|--------|
| [6] Mobility Support with Intelligent Obstacle Detection for Enhanced Safety                                    | Yes                   | No                 | No               | Yes            | Medium |
| [5] Smart Stick Navigation System for Visually Impaired Based on Machine Learning Algorithms Using Sensors Data | Yes                   | Yes                | No               | Yes            | High   |
| [7] A Smart Cane Based on 2D LiDAR and RGB-D Camera Sensor Realizing Navigation and Obstacle Recognition        | Yes                   | No                 | No               | Yes            | High   |
| Proposed SafeStep                                                                                               | Yes                   | Yes                | Yes              | Yes            | Low    |

## **2.4 Research Gap / Limitations of Existing Methods**

From the literature review we were able to identify many traditional similarities but the main limitations for this sector were also clearly identified such as having high implementation costs, limited hazard detection capabilities, lack of integrated safety mechanisms, poor affordability for developing countries, limited emergency communication support and lastly heavy dependence on advanced sensors.

But our proposed SafeStep system addresses these gaps through low-cost architecture, multi-sensor obstacle detection, water hazard identification, IoT-enabled SOS alerts, lightweight design and affordable deployment model.

## **2.5 Summary**

This chapter reviewed the theoretical foundations of assistive robotics, embedded systems, ultrasonic sensing, and IoT technologies. Existing studies demonstrate the effectiveness of intelligent mobility aids but reveal several limitations regarding affordability, hazard detection, and integrated safety functionality. The identified research gaps justify the development of SafeStep as a low-cost, multi-functional smart walker capable of providing obstacle avoidance, water hazard detection, and emergency communication assistance within a single platform.

# Chapter 3

## System Analysis and Design

This chapter presents the requirements analysis, system architecture, design considerations, and algorithmic workflow of the proposed SafeStep smart walker. The objective is to establish a comprehensive understanding of how various hardware and software components interact to provide intelligent mobility assistance for blind and elderly users.

### 3.1 Requirement Analysis

The system requirements were identified through analysis of mobility challenges faced by elderly individuals and visually impaired users, observations of conventional mobility aids, and review of related assistive navigation systems.

#### 3.1.1 Stakeholder Analysis

The primary stakeholders include:

##### Primary Users

- Visually impaired individuals
- Elderly individuals
- Users with limited mobility

## Secondary Users

- Family members
- Caregivers
- Healthcare assistants

## Technical Stakeholders

- System developers
- Maintenance personnel
- Future researchers

### 3.1.2 Functional Requirements

Table 3.1: Functional Requirements

| ID   | Requirement Description                         |
|------|-------------------------------------------------|
| FR1  | Detect obstacles in front of the walker         |
| FR2  | Detect obstacles on the left side               |
| FR3  | Detect obstacles on the right side              |
| FR4  | Automatically select safer navigation direction |
| FR5  | Detect wet or slippery surfaces                 |
| FR6  | Stop movement upon water hazard detection       |
| FR7  | Generate directional buzzer alerts              |
| FR8  | Send SOS notifications through Blynk            |
| FR9  | Enable ON/OFF control through push button       |
| FR10 | Provide semi-autonomous navigation support      |

### 3.1.3 Non-Functional Requirements

Table 3.2: Non-Functional Requirements

| ID   | Requirement Description         |
|------|---------------------------------|
| NFR1 | Low-cost implementation         |
| NFR2 | Lightweight construction        |
| NFR3 | Response time below 1 second    |
| NFR4 | Reliable wireless communication |
| NFR5 | User-friendly operation         |
| NFR6 | Low power consumption           |
| NFR7 | Easy maintenance                |
| NFR8 | Expandable architecture         |

## 3.2 System Overview / Architecture

SafeStep follows a layered architecture consisting of sensing, processing, communication, and actuation layers.



### 3.2.1 High-Level Architecture

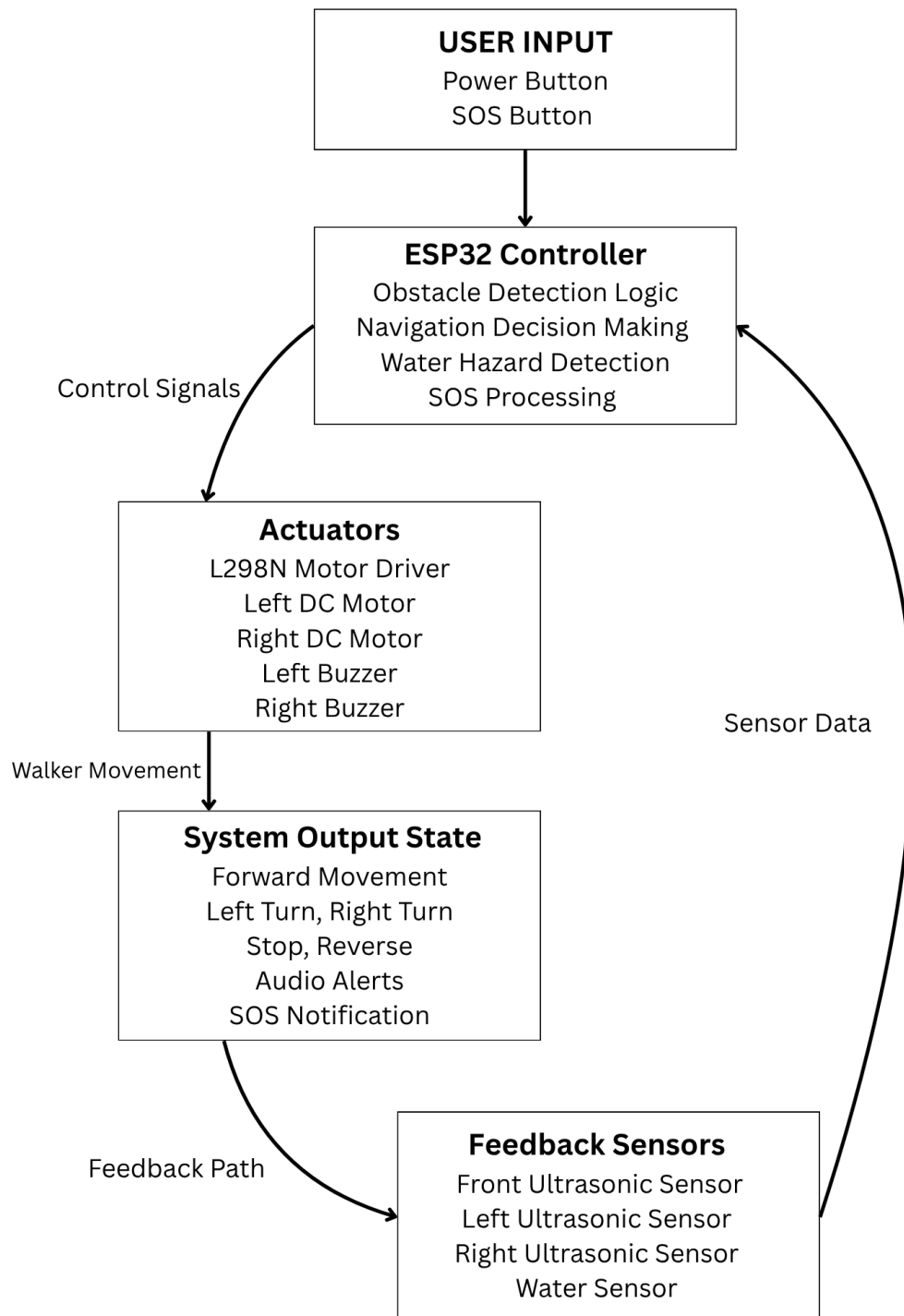


Figure 3.1: SafeStep Control System Diagram

### 3.2.2 Detailed Design Diagrams

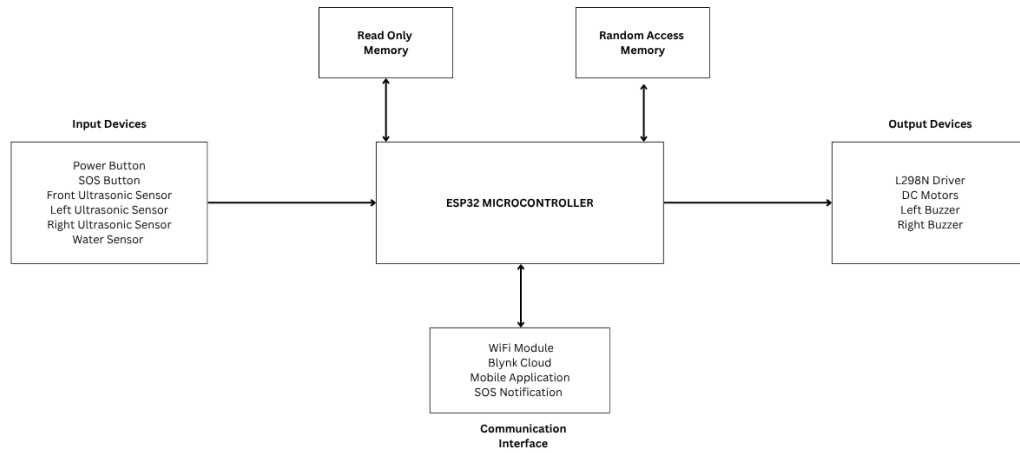


Figure 3.2: Hardware Achitecture of SafeStep Embedded System

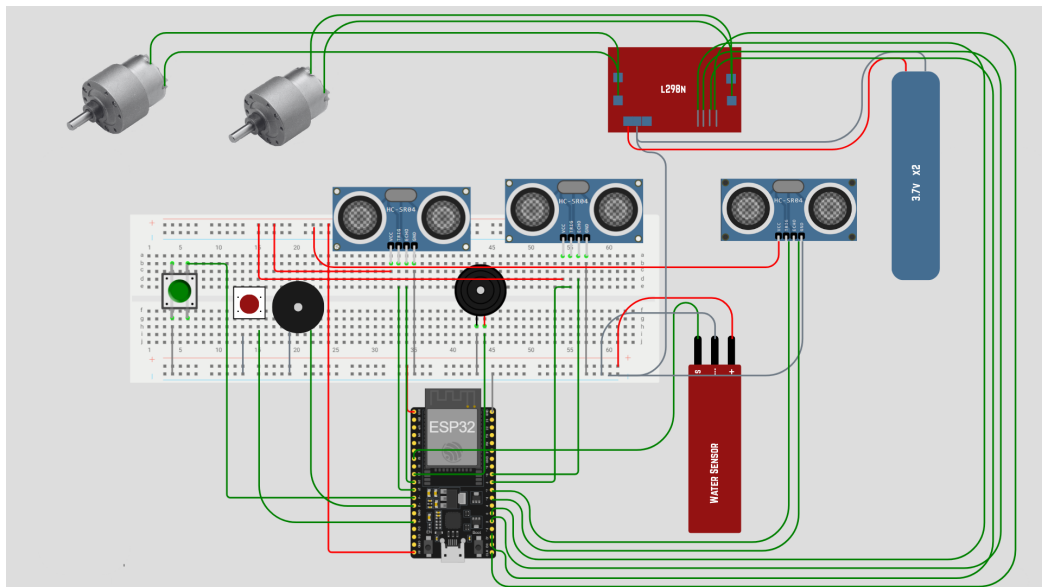


Figure 3.3: Circuit Diagram

### 3.2.3 Design Alternatives and Rationale

Several design alternatives were evaluated before selecting the final architecture.

Table 3.3: Obstacle Detection Technologies

| Technology        | Cost      | Complexity |
|-------------------|-----------|------------|
| Camera            | High      | High       |
| LiDAR             | Very High | High       |
| Ultrasonic Sensor | Low       | Low        |

Ultrasonic sensing was selected due to affordability and simplicity.

Table 3.4: Controller Selection

| Controller   | Cost   | WiFi |
|--------------|--------|------|
| Arduino Uno  | Medium | No   |
| Raspberry Pi | High   | Yes  |
| ESP32        | Low    | Yes  |

ESP32 was selected due to having a integrated WiFi, low cost, sufficient GPIO pins, low power consumption.

### 3.3 Algorithmic Flow and Pseudocode

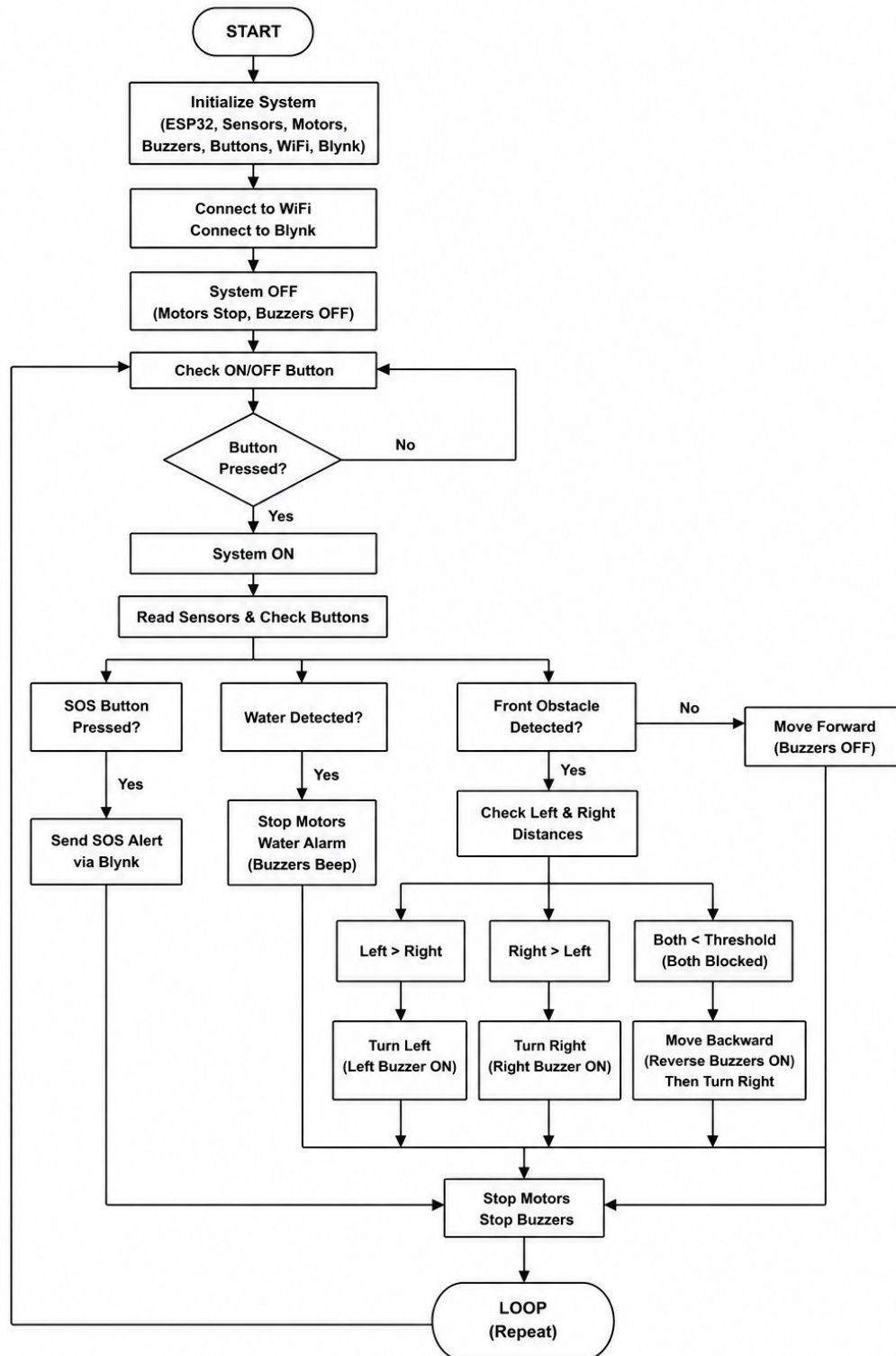


Figure 3.4: SafeStep Flowchart

The navigation algorithm continuously monitors environmental conditions and determines appropriate actions.

## **Pseudocode**

INITIALIZE esp32

INITIALIZE sensors

INITIALIZE blynk

WHILE system running

    CHECK SOS button

    IF SOS pressed

        SEND blynk notification

    ENDIF

    CHECK main power button

    IF OFF

        STOP motors

        STOP buzzers

        continue

    ENDIF

    READ water sensor

    IF water DETECTED

        STOP motors

        ACTIVATE water alarm

```
        continue

ENDIF

READ front distance

IF front clear

    MOVE forward

ELSE

    STOP motors

    READ left distance

    READ right distance

    IF both blocked

        MOVE backward

        TURN right

    ELSE IF left clear

        TURN left

    ELSE

        TURN right

    ENDIF

ENDIF

ENDIF
```

## Navigation Decision Function

The walker selects navigation direction based on:

$$Direction = \begin{cases} Left & \text{if } left \text{ is clear} \\ Right & \text{if } right \text{ is clear} \\ Reverse & \text{otherwise} \end{cases} \quad (3.1)$$

### 3.4 Summary

This chapter presented the requirements analysis, system architecture, hardware design, and navigation algorithm of the SafeStep smart walker. The modular architecture enables efficient interaction among sensing, processing, communication, and actuation modules. The design emphasizes affordability, simplicity, and user safety while maintaining expandability for future enhancements.

# Chapter 4

## Methodology and Implementation

This chapter describes the development methodology, hardware and software technologies, implementation procedures, and system integration process used to construct the SafeStep smart walker.

### 4.1 Methodological Framework

The project followed a structured engineering development methodology consisting of seven phases.

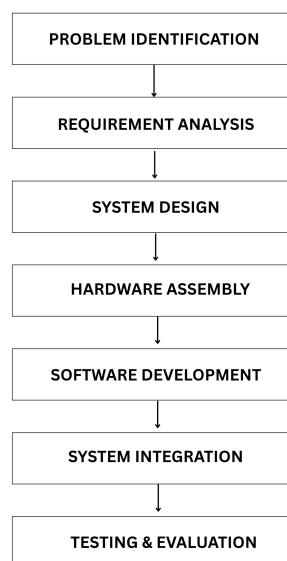


Figure 4.1: Project Development Framework



## 4.2 Tools, Libraries, and Technologies Used

Table 4.1: Development Technologies

| Category             | Technology                |
|----------------------|---------------------------|
| Programming Language | C++                       |
| IDE                  | Arduino IDE               |
| Microcontroller      | ESP32                     |
| Cloud Platform       | Blynk                     |
| Communication        | WiFi                      |
| Sensor Technology    | Ultrasonic + Water Sensor |
| Motor Control        | L298N                     |

### Software Libraries

Table 4.2: Libraries Used

| Library            | Purpose                  |
|--------------------|--------------------------|
| WiFi.h             | WiFi communication       |
| WiFiClient.h       | Client networking        |
| BlynkSimpleEsp32.h | IoT notification support |

## 4.3 Data Collection / Dataset Description

Since this is an IoT and robotics project, data was collected through sensors rather than datasets.

### Sensor Data Types

- Front Ultrasonic sensor gave us Front distance

- Left Ultrasonic sensor gave us Left distance
- Right Ultrasonic sensor gave us Right distance
- Water sensor gave us Moisture level
- Buttons gave us User input

## 4.4 Preprocessing Steps

The following preprocessing operations were implemented to get a better and acceptable output : noise filtering through repeated sensing, distance threshold comparison, water sensor threshold calibration, debounce handling for buttons, invalid sensor reading rejection.

### Threshold Values

Table 4.3: Threshold Values

| Parameter         | Value |
|-------------------|-------|
| Obstacle Distance | 20 cm |
| Water Threshold   | 200   |
| Button Debounce   | 50 ms |

## 4.5 Model Development / Algorithm Design

Unlike machine learning systems, SafeStep utilizes a rule-based decision model.

### Decision Logic

1. When Front is Clear, Move Forward
2. When Obstacle is Detected, Check Side Sensors

3. When Left is More Open, Turn Left
4. When Right is More Open, Turn Right
5. When Both are Blocked then Reverse + Turn
6. When Water is Detected, Stop
7. When SOS is Pressed, Send Alert

## 4.6 System Implementation / Modules Description

The implementation consists of five major modules.

**Module 1: Obstacle Detection Module** Uses three HC-SR04 sensors to measure surrounding distances. Functions: Front scanning, Left scanning, Right scanning.

**Module 2: Navigation Module** Determines movement direction. Capabilities: Forward movement, Left turn, Right turn, Reverse recovery.

**Module 3: Water Hazard Module** Monitors moisture levels. Functions: Surface analysis, Hazard detection, Movement stopping

**Module 4: Audio Feedback Module** Provides directional information. Conditions and their buzzers : When Left Turn, Left Buzzer is activated and When Right Turn, Right Buzzer is activated and While reversing, Both Buzzers are activated and When Water is detected Both Buzzers give Pulses.

**Module 5: IoT Emergency Module** Features: WiFi connectivity, Blynk integration, SOS notification

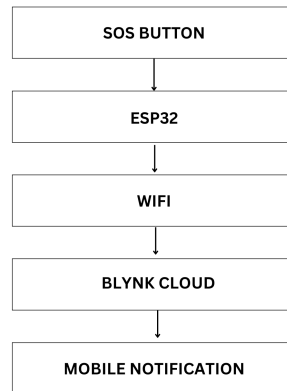


Figure 4.2: Blynk Communication Workflow

## 4.7 Hardware and Software Requirements

### Hardware Requirements

Table 4.4: Hardware Components

| Component      | Quantity |
|----------------|----------|
| ESP32          | 1        |
| HC-SR04        | 3        |
| Water Sensor   | 1        |
| L298N          | 1        |
| DC Motor       | 2        |
| Buzzer         | 2        |
| Push Button    | 2        |
| Battery        | 2        |
| Walker Chassis | 1        |

## Software Requirements

Table 4.5: Software Requirements

| Software            | Version       |
|---------------------|---------------|
| Arduino IDE         | 2.x           |
| ESP32 Board Package | Latest        |
| Blynk IoT Platform  | Latest        |
| Windows OS          | Windows 10/11 |

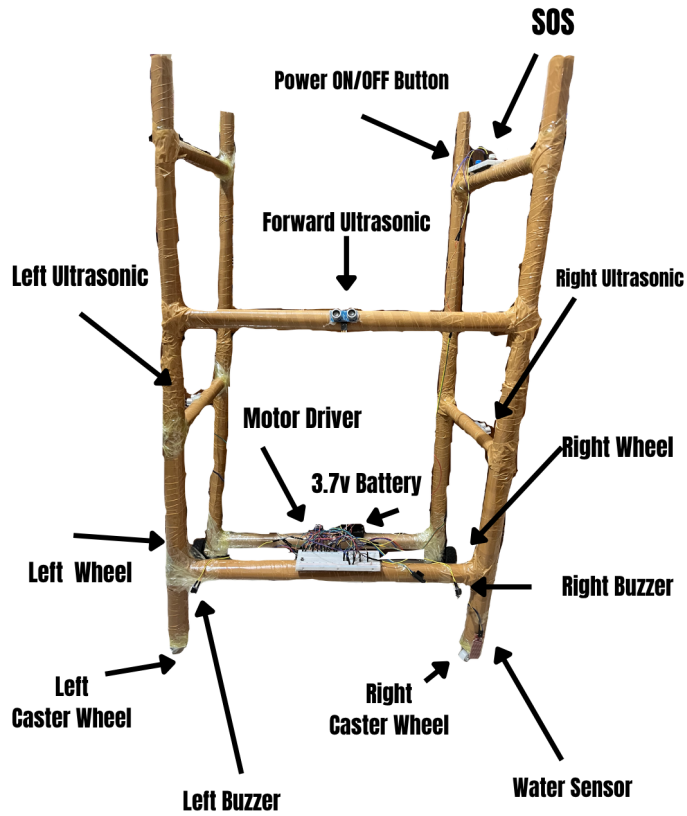


Figure 4.3: Prototype Photograph

## 4.8 Summary

This chapter described the development methodology, implementation process, hardware architecture, software framework, and system modules of the SafeStep smart walker. The integration of sensing, navigation, IoT communication, and safety mechanisms establishes a complete assistive mobility platform suitable for elderly and visually impaired users.

# Chapter 5

## Results and Evaluation

This chapter presents the experimental evaluation and performance analysis of the SafeStep smart walker prototype. Multiple testing scenarios were conducted to assess obstacle detection, navigation assistance, water hazard detection, emergency notification, and overall system responsiveness. The obtained results demonstrate the effectiveness and practicality of the proposed assistive navigation system.

### 5.1 Experimental Setup

The prototype was physically assembled and tested under indoor and semi-controlled environmental conditions.

The evaluation focused on: Obstacle detection capability, Directional navigation assistance, Water hazard detection, SOS emergency notification, System response time, Battery performance, Hardware Setup.

The experimental setup consisted of: ESP32 microcontroller, Three HC-SR04 ultrasonic sensors, Water detection sensor, L298N motor driver, Two DC motors, Two buzzers, Two push buttons, Blynk mobile application, Dual 3.7V battery supply.

### 5.1.1 Testing Scenarios

Table 5.1: Test Scenarios

| Test Type            | Number of Tests |
|----------------------|-----------------|
| Obstacle Detection   | 50              |
| Navigation / Turning | 30              |
| Water Detection      | 20              |
| SOS Notification     | 20              |
| Total Tests          | <b>120</b>      |

## 5.2 Performance Metrics

The following performance metrics were used. **Obstacle Detection Accuracy**

$$Accuracy = \frac{CorrectDetections}{TotalTests} \times 100 \quad (5.1)$$

### Water Detection Reliability

$$Reliability = \frac{SuccessfulWaterDetections}{TotalWaterTests} \times 100 \quad (5.2)$$

### Notification Delay

Average time required for notification delivery.

### System Response Time

Time between obstacle detection and navigation decision.



## 5.3 Result Analysis

### 5.3.1 Obstacle Detection Performance

The obstacle detection module achieved strong performance across different obstacle positions and distances.

Table 5.2: Obstacle Detection Results

| Parameter             | Value      |
|-----------------------|------------|
| Total Tests           | 50         |
| Successful Detections | 45         |
| Detection Accuracy    | 90%        |
| Detection Range       | 5–150 cm   |
| Optimal Range         | 20–80 cm   |
| Response Time         | < 1 second |

The results indicate that ultrasonic sensing provides sufficient environmental awareness for assistive navigation applications.



Figure 5.1: Obstacle Detection Accuracy Chart

### 5.3.2 Water Hazard Detection Performance

The water detection module was evaluated using multiple wet surface scenarios.

Table 5.3: Water Detection Results

| Parameter             | Value |
|-----------------------|-------|
| Tests Conducted       | 20    |
| Successful Detections | 14    |
| Reliability           | 70%   |

The lower accuracy compared to obstacle detection can be attributed to sensor sensitivity variations and environmental moisture conditions.

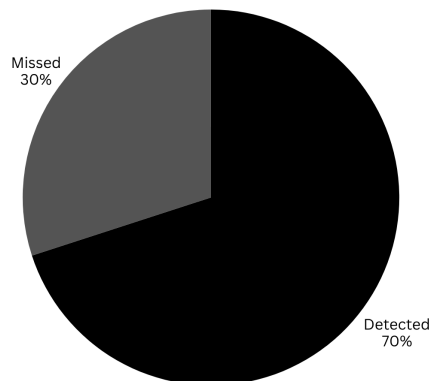


Figure 5.2: Water Detection Reliability

### 5.3.3 SOS Notification Performance

The emergency communication feature demonstrated reliable operation.

Table 5.4: SOS Notification Results

| Parameter                | Value       |
|--------------------------|-------------|
| Total Tests              | 20          |
| Successful Notifications | 20          |
| Success Rate             | 100%        |
| Average Delay            | 1–2 Seconds |

The Blynk cloud platform successfully delivered notifications with minimal latency.

### 5.3.4 Navigation Performance

The walker was tested in various obstacle arrangements.

Table 5.5: Navigation Evaluation

| Scenario        | Performance                 |
|-----------------|-----------------------------|
| Straight Path   | Successful                  |
| Left Obstacle   | Turn Right                  |
| Right Obstacle  | Turn Left                   |
| Both Sides Open | Chosen Safest Route         |
| Front Blocked   | Avoidance Successful        |
| Fully Blocked   | Reverse Recovery Successful |

The navigation algorithm successfully performed semi-autonomous movement decisions.

### 5.3.5 Battery Performance

Table 5.6: Battery Analysis

| Parameter       | Value                    |
|-----------------|--------------------------|
| Battery Type    | $2 \times 3.7V$          |
| Configuration   | Direct Supply            |
| Runtime         | Approximately 20 Minutes |
| Motor Condition | Underpowered Operation   |

The limited battery runtime represents one of the primary constraints of the current prototype.

## 5.4 Comparison with Existing Approaches

A comparison was conducted against representative assistive navigation systems reported in recent literature.

Table 5.7: Comparative Analysis

| Feature                    | Smart Cane | Smart Stick | Advanced Walker | SafeStep |
|----------------------------|------------|-------------|-----------------|----------|
| Obstacle Detection         | Yes        | Yes         | Yes             | Yes      |
| Water Detection            | No         | Limited     | Limited         | Yes      |
| SOS Alert                  | No         | Yes         | Yes             | Yes      |
| IoT Support                | No         | Yes         | Yes             | Yes      |
| Semi-Autonomous Navigation | No         | No          | Yes             | Yes      |
| Low Cost                   | Moderate   | Moderate    | No              | Yes      |

The comparison demonstrates that SafeStep integrates multiple safety functionalities while maintaining affordability.

## 5.5 Discussion on Findings

The experimental results demonstrate that the proposed SafeStep system effectively addresses several mobility challenges faced by elderly and visually impaired users. Key findings include: High obstacle detection accuracy (90%), Fast reaction time (<1 second), Reliable emergency notification capability, Effective directional guidance, Low implementation cost, Lightweight architecture.

The integration of multiple assistive functionalities into a single platform distinguishes SafeStep from many existing low-cost navigation aids.

The results also validate the feasibility of utilizing low-cost embedded systems and IoT technologies for assistive healthcare applications in developing countries.

## 5.6 Limitations

Despite encouraging results, several limitations were observed. Few Hardware Limitations were observed such as Limited battery capacity, underpowered motor operation, dependence on WiFi availability. Also few Sensor Limitations were also visible such as ultrasonic blind spots, reduced water detection reliability, environmental noise sensitivity. And the future works may include GPS tracking, computer vision, object classification, No voice interaction. These limitations provide opportunities for future improvements.

## 5.7 Summary

This chapter evaluated the performance of the SafeStep prototype through multiple experimental scenarios. Results demonstrated 90% obstacle detection accuracy, 70%

water hazard detection reliability, and successful emergency communication capabilities. The findings confirm the viability of SafeStep as a low-cost assistive mobility platform.

## Chapter 6

# Project Planning and Budget

This chapter presents the project timeline, resource allocation strategy, and budget estimation associated with the development of the SafeStep smart walker.

### 6.1 Project Timeline

A structured development approach was followed throughout the project lifecycle.

The project was completed over seven weeks.

Table 6.1: Seven-Week Project Timeline

| Activities            | W1 | W2 | W3 | W4 | W5 | W6 | W7 | W8 |
|-----------------------|----|----|----|----|----|----|----|----|
| Literature Review     |    |    |    |    |    |    |    |    |
| Requirement Analysis  |    |    |    |    |    |    |    |    |
| System Design         |    |    |    |    |    |    |    |    |
| Hardware Procurement  |    |    |    |    |    |    |    |    |
| Circuit Development   |    |    |    |    |    |    |    |    |
| Software Development  |    |    |    |    |    |    |    |    |
| Prototype Integration |    |    |    |    |    |    |    |    |
| Testing & Validation  |    |    |    |    |    |    |    |    |
| Documentation         |    |    |    |    |    |    |    |    |
| Final Presentation    |    |    |    |    |    |    |    |    |

## 6.2 Budget Estimation

The total project budget was kept intentionally low to maintain affordability and accessibility.



Table 6.2: Hardware Cost Breakdown

| Component                   | Quantity | Unit Cost (BDT) | Total Cost (BDT) |
|-----------------------------|----------|-----------------|------------------|
| ESP32                       | 1        | 600             | 600              |
| HC-SR04 Sensor              | 3        | 100             | 300              |
| L298N Driver                | 1        | 300             | 300              |
| DC Motors                   | 2        | 500             | 1000             |
| 3.7V Batteries              | 2        | 200             | 400              |
| Buzzers                     | 2        | 100             | 200              |
| Walker Chassis Materials    | 1        | 500             | 500              |
| Miscellaneous Components    | –        | 1000            | 1000             |
| <b>Total Estimated Cost</b> |          |                 | <b>4300 BDT</b>  |

Table 6.3: Cost Distribution

| Category      | Percentage |
|---------------|------------|
| Motors        | 23.3%      |
| Electronics   | 32.6%      |
| Power System  | 9.3%       |
| Chassis       | 11.6%      |
| Miscellaneous | 23.2%      |

### 6.2.1 Cost-Benefit Analysis

Compared with commercial smart mobility aids that often cost between 20,000 and 100,000+ BDT, the SafeStep prototype was developed for approximately: **4300 BDT**

This represents a significantly lower implementation cost while maintaining core as-

sistive functionalities.

Benefits include:

- Affordable deployment
- Simple maintenance
- Easy hardware replacement
- Scalability for future enhancements

## **6.3 Summary**

This chapter presented the project timeline and budget estimation associated with the development of SafeStep. The seven-week development schedule ensured systematic project execution, while the low-cost budget of approximately 4300 BDT demonstrated the feasibility of creating affordable assistive navigation systems for developing regions.

# Chapter 7

## Standards and Design Constraints

This chapter discusses the engineering standards, professional practices, design constraints, ethical considerations, sustainability concerns, and accreditation requirements associated with the SafeStep smart walker project. Furthermore, the chapter demonstrates compliance with Complex Engineering Problem (CEP) and Complex Engineering Activity (CEA) requirements commonly used in engineering accreditation frameworks.

### 7.1 Compliance with Standards and Professional Practice

The development of SafeStep followed established engineering principles related to embedded systems, IoT communication, assistive technology, and user safety.

The project emphasizes: Reliable sensing and control, User safety, Ethical use of technology, Low-power system design, Accessibility for vulnerable populations, Maintainable system architecture.

#### 7.1.1 Software Standard

The software architecture follows embedded programming best practices.

Applicable standards include: IEEE Software Engineering Code of Ethics, IEEE Stan-

dard for Software Verification and Validation, Secure IoT communication principles, Modular programming practices.

Key software compliance features: Structured code organization, Modular functions, Input validation, Debounce protection, Fault handling through sensor timeout management.

### 7.1.2 Hardware Standard

The hardware implementation follows standard embedded system design principles.

Relevant standards include IEEE Standards Association recommendations, IEC safety considerations for electronic systems, Standard HC-SR04 operating specifications, ESP32 manufacturer design guidelines.

Hardware compliance considerations Safe operating voltage, Proper GPIO utilization, Controlled motor actuation, Sensor calibration procedures.

### 7.1.3 Assistive Technology Considerations

The system was designed with accessibility as a primary objective.

Design principles include Simple user interaction, Minimal training requirements, Audible feedback mechanisms, Emergency assistance support, Lightweight construction.

## 7.2 Design Constraints

Several constraints influenced the final design of the SafeStep system. **Power Constraints**

The prototype operates using Two 3.7V batteries.

This resulted in Limited runtime, Reduced motor torque, Lower navigation speed.

**Sensor Constraints**

Ultrasonic sensors introduce Blind zones, Surface reflection issues, Environmental sensitivity.

### **Communication Constraints**

The SOS module requires Internet access, WiFi connectivity, Blynk cloud availability.

### **Budget Constraints**

The project was designed under strict affordability requirements.

Total budget: 4300 BDT

This significantly influenced component selection decisions.

## **7.2.1 Ethical and Professional Responsibilities**

**Key Ethical Considerations include:**

### **User Safety**

The system prioritizes Hazard avoidance, Emergency communication, Fall prevention support.

### **Honest Reporting**

All experimental results presented in this report represent actual testing outcomes obtained during prototype evaluation.

### **Accessibility**

The project was specifically designed to benefit:

- Elderly individuals
- Visually impaired users

### **Intellectual Property**

All third-party software libraries and technologies are properly acknowledged.

## 7.2.2 Environmental and Sustainability Considerations

Sustainability was considered throughout system development.

**Positive Environmental Contributions** are Low power embedded hardware, Reusable electronic components, Minimal computational requirements, Small electronic footprint.

**Resource Optimization** The system utilizes Efficient ESP32 processing, Lightweight sensors, Cloud services only when necessary.

## 7.2.3 Health and Safety Considerations

The project directly contributes to mobility safety.

Potential hazards addressed include:

### Obstacle Collisions

Mitigation:

- Multi-direction obstacle detection

### Slippery Surfaces

Mitigation:

- Water hazard sensing

### Emergency Situations

Mitigation:

- SOS notification functionality

## Navigation Errors

Mitigation:

- Directional buzzer guidance

## 7.3 Complex Engineering Problem (CEP) Coverage

The SafeStep project addresses several attributes associated with Complex Engineering Problems. **CEP Mapping**

Table 7.1: CEP Mapping

| CEP Attribute                        | How Addressed in Project                    | Evidence       |
|--------------------------------------|---------------------------------------------|----------------|
| P1: Depth of Knowledge               | Embedded systems, IoT, robotics integration | Chapters 3 & 4 |
| P2: Conflicting Requirements         | Cost vs functionality trade-offs            | Section 3.2.3  |
| P3: In-depth Analysis                | Sensor selection and navigation logic       | Chapter 3      |
| P4: Familiar and Unfamiliar Problems | Mobility assistance challenges              | Chapters 1 & 2 |
| P5: Wide-Ranging Issues              | Safety, accessibility, affordability        | Chapter 7      |
| P6: Stakeholder Needs                | Elderly and visually impaired users         | Section 3.1    |
| P7: Significant Consequences         | User safety and mobility assistance         | Chapter 5      |

### CEP Justification

The project required multidisciplinary knowledge involving:

- Robotics
- Embedded systems
- Electronics
- IoT communication
- Human-centered design

The integration of these domains demonstrates the complexity expected in engineering problem solving.

## 7.4 Complex Engineering Activities (CEA) Coverage

The project also satisfies several Complex Engineering Activity requirements. **CEA Mapping**

Table 7.2: CEA Mapping

| CEA Attribute                  | How Addressed                            | Evidence       |
|--------------------------------|------------------------------------------|----------------|
| A1: Diverse Resources          | Hardware, software, cloud services       | Chapters 3 & 4 |
| A2: Resolution of Interactions | Sensor-actuator coordination             | Chapter 4      |
| A3: Creative Innovation        | Combined obstacle, water, and SOS system | Chapter 2      |
| A4: Consequences to Society    | Mobility assistance and safety           | Chapter 1      |
| A5: Engineering Judgment       | Component selection and optimization     | Chapter 3      |



## 7.5 Summary

This chapter demonstrated compliance with engineering standards, ethical responsibilities, sustainability considerations, and accreditation requirements. The CEP and CEA mappings confirm that SafeStep addresses a complex engineering problem through the application of multidisciplinary engineering knowledge and innovative system development.

# Chapter 8

## Conclusion and Future Work

This chapter summarizes the major outcomes of the SafeStep project, highlights key contributions, discusses limitations, and outlines future research opportunities.

### 8.1 Summary of the Work

Restates objectives and summarizes the project achievements. This project presented the design, implementation, and evaluation of SafeStep, a low-cost IoT-enabled semi-autonomous smart walker intended to assist elderly and visually impaired users during navigation. The system integrates ESP32 microcontroller, Three ultrasonic sensors, Water hazard sensor, Dual directional buzzers, Motorized navigation support, Blynk-based SOS communication.

The developed prototype demonstrates the feasibility of combining embedded systems, sensing technologies, and IoT communication into a practical assistive mobility solution.

### 8.2 Key Findings and Contributions

The primary contributions of this research include:

#### **C1: Multi-Hazard Detection**

The system detects Obstacles, Water hazards, Emergency conditions within a unified architecture.

### **C2: Affordable Design**

The complete prototype was developed at approximately: 4300 BDT making it significantly cheaper than many commercial alternatives.

### **C3: IoT Integration**

The SOS functionality successfully enables Real-time emergency notification, Care-giver awareness, Remote assistance.

### **C4: Semi-Autonomous Navigation**

The walker demonstrates Automatic obstacle avoidance, Directional decision making, Safe route selection.

**C5: Real-World Validation** The prototype was physically built and experimentally tested. Key results include:

- 90% obstacle detection accuracy
- 70% water detection reliability
- < 1 second response time
- 1–2 second SOS notification delay

## **8.3 Limitations of the Study**

Despite promising results, several limitations remain. Hardware Limitations are Low battery runtime, Underpowered motors, Lightweight prototype frame and Sensor Limitations are Ultrasonic blind zones, Environmental interference. Lastly Functional Limitations are No GPS functionality, No voice assistance, No object recognition, No machine learning capability.

## 8.4 Recommendations and Future Work

Several improvements can enhance future versions of SafeStep.

### **GPS Tracking**

Integration of GPS modules can enable Real-time location tracking, Route monitoring, Emergency location sharing.

**AI-Based Object Recognition** Future systems may incorporate Computer vision, Object classification, Scene understanding using lightweight AI models.

**Voice Assistance** Voice interaction can improve usability through Audio navigation instructions, Voice commands, Emergency voice alerts.

### **Advanced Power Management**

Future enhancements may include Rechargeable battery packs, Solar-assisted charging, Battery health monitoring.

### **Mobile Application Development**

A dedicated mobile application can provide Live walker status, Navigation history, Remote caregiver monitoring.

### **Healthcare Integration**

Potential future features include Heart rate monitoring, Fall detection, Health analytics.

# References

- [1] Isaac Asante, Lau Bee Theng, Mark Tee Kit Tsun, and John Chua. Ecofriendly autonomous smart walker for enhanced ambulatory assistance. IEEE Potentials, 2024.
- [2] Various Authors. idog: An intelligent guide dog harness for visually impaired pedestrians based on artificial intelligence and edge computing. IEEE Sensors Journal, 24(24):41997–42008, 2024.
- [3] Various Authors. A smart clothing approach for augmenting mobility of visually impaired people. IEEE Access, 12:24659–24671, 2024.
- [4] Various Authors. Smart walking cane based on triboelectric nanogenerators for assisting the visually impaired. Nano Energy, 124:109485, 2024.
- [5] Sadik Kamel Gharghan, Hussein S. Kamel, Asaower Ahmad Marir, and Lina Akram Saleh. Smart stick navigation system for visually impaired based on machine learning algorithms using sensors data. Journal of Sensor and Actuator Networks, 13(4):43, 2024.
- [6] Jong Hyeok Han, Inkwon Yoon, Hyun Soo Kim, et al. Mobility support with intelligent obstacle detection for enhanced safety. Optics, 5(4):434–444, 2024.
- [7] Chunming Mai, Huaze Chen, Lina Zeng, et al. A smart cane based on 2d lidar and rgb-d camera sensor realizing navigation and obstacle recognition. Sensors, 24(3):870, 2024.
- [8] Usman Masud, Tareq Saeed, and Ghulam Abbas. Smart assistive system for visually impaired people obstruction avoidance through object detection and classification. IEEE Access, 10:13428–13441, 2022.

- [9] Gabriel Iluebe Okolo, Turke Althobaiti, and Naeem Ramzan. Assistive systems for visually impaired persons: Challenges and opportunities for navigation assistance. Sensors, 24(11):3572, 2024.
- [10] Baoyi Zhang, Zongsheng Wang, and Zhe Li. Mobility aid design for the elderly (made): A design thinking approach using a smart walker as a case study. Humanities and Social Sciences Communications, 11:1469, 2024.

# Appendices

## A Source Code / GitHub Link

GitHub Repository: <https://github.com/MD-Sydul-Islam-Zubair/Smart-Walker>

## B Team Responsibilities

### MD. Sydul Islam Zubair

- Complete software development
- ESP32 programming
- Blynk integration
- Motor driver interfacing
- Hardware integration
- Project supervision

### Suna Hera Akter Suna

- Documentation support
- Testing assistance
- Literature review support
- Chassis creation

**Farhana Haque Mahima**

- Conducted Survey 1
- Conducted Survey 2

## **C Ethics and Compliance Statements**

The authors declare that:

- Experimental results reported are based on actual prototype testing.
- No human subject data was collected.
- The project was developed for educational and research purposes.
- All third-party resources are properly acknowledged.

## **D Survey 1**

**SQ1:** Your Name

**Ans:** Delowara begum.

**SQ2:** What is your job role?

**Ans:** Retired.

**SQ3:** Are you involved in purchasing/decision making?

**Ans:** No.

**SQ4:** Have you understood the product concept?

**Ans:** 5 out of 5.

**SQ5:** Do you think our product will be helpful for your workplace?

**Ans:** 5 out of 5.



**Additional comments:**

This is a very good initiative for our elderly people.

## **E Survey 2**

**SQ1:** Your Name

**Ans:** Jamila Khatun.

**SQ2:** What is your job role?

**Ans:** Retired.

**SQ3:** Are you involved in purchasing/decision making?

**Ans:** No.

**SQ4:** Have you understood the product concept?

**Ans:** 5 out of 5.

**SQ5:** Do you think our product will be helpful for your workplace?

**Ans:** 4 out of 5.

**SQ6:** Would you purchase this product?

**Ans:** 3 out of 5.

**SQ7** Do you think the product price is acceptable?

**Ans:** 4 out of 5.

**SQ8:** How much money would you spend for this ind of product?

**Ans:** From 3000BDT- 5000BDT

**SQ9:** Which part of this product needs improvement?

**Ans:** The battery life should be improved because it runs out quite quickly from what I heard. The buzzer volume could also be adjustable for users with hearing difficulties.

**Additional comments:**

The walker will help me notice objects that I might not have seen otherwise. The emer-

gency notification feature is also reassuring because family members can be alerted when needed. Overall, it felt useful and practical.



Figure 1: Collected Images while conducting survey 2